F7AB - Actuarial and Financial Mathematics
Tutorial 10 – Term structure of interest rates

Assume that time is measured in years. Let

$$\begin{aligned} f_{t,n} &= n\text{-year forward rate for an investment made at time } t \text{ for } n \text{ years.} \\ y_n &= \text{spot rate from time 0 to time } n. \\ c_n &= \text{par yield from time 0 to time } n. \\ P(n) &= \text{price at time 0 of a zero coupon bond that matures at time } n. \end{aligned}$$

Question 1 You are given that

$$\begin{aligned} f_{0,1} &= 6.0\% \text{ p.a. effective,} \\ f_{0,2} &= 6.5\% \text{ p.a. effective, and} \\ f_{1,2} &= 6.6\% \text{ p.a. effective.} \end{aligned}$$

Determine the 3-year par yield.

Question 2 The following n-year spot rates p.a. effective were observed at time $t = 0$.

n	1	2	3	4	5	6
y_n	4%	5%	6%	7%	7.5%	8%

1. Define what is meant by an n-year spot rate.
2. Calculate $f_{3,2}$.
3. Using the above n-year spot rates, calculate the 6-year par yield at time $t = 0$.

Question 3 You are given

$$\begin{aligned} f_{0,1} &= 6.0\% \text{ p.a. effective.} \\ f_{0,2} &= 6.5\% \text{ p.a. effective.} \\ f_{1,2} &= 6.6\% \text{ p.a. effective.} \end{aligned}$$

1. Calculate the 2-year spot rate.
2. Calculate the 3-year spot rate.

Question 4 You are given the yield to redemption on a n-year bond with a 6% coupon payable annually, for $n = 1, 2, 3$ as follows.

n	1	2	3
Yield to redemption (p.a. effective)	6%	6.3%	6.6%

All bonds are redeemed at par (i.e. the redemption proceeds equal 1) and are exactly one year from the next coupon payment.

1. Calculate y_1, y_2, and y_3 assuming no arbitrage.
2. Calculate $f_{0,1}$, $f_{1,1}$, and $f_{2,1}$ assuming no arbitrage.
3. Explain why the forward rates above increase more rapidly with term than the spot rates.

Question 5 You are given the following information about par yields (p.a. effective).

n	1	2	3	4	5	6	7	8
c_n	0.060	0.057	0.056	0.058	0.062	0.064	0.064	0.064

Calculate the values of $P(n)$ and y_n for $n = 1, 2, \ldots, 8$.

Question 6 Denote by $F(0, t, T)$ the forward rate (p.a. effective) at time 0 that applies between times t and T.

You are given the following information about par yields.

n	1	2	3
c_n	0.05263	0.05405	0.03261

1. Calculate the values of $P(n)$ for $n = 1, 2, 3$.
2. Hence derive the forward rates $F(0, 1, 2)$, $F(0, 1, 3)$ and $F(0, 2, 3)$.
3. Comment on the arbitrage possibilities presented by the specified values of the par-yield curve.

Answers: Q1. 6.395% p.a.; Q2.2. 9.790% p.a.; Q2.3. 7.708% p.a.; Q3.1. 6.5% p.a.; Q3.2. 6.400% p.a.; Q4.1. $y_1 = 6\%$ p.a, $y_2 = 6.309\%$ p.a, $y_3 = 6.625\%$ p.a.; Q4.2. $f_{0,1} = 6\%$ p.a., $f_{1,1} = 6.619\%$ p.a, $f_{2,1} = 7.259\%$ p.a.; Q5. Some sample solutions: $P(1) = 0.943396$ and $y_1 = 0.06$, $P(2) = 0.895200$ and $y_2 = 0.056915$, $P(5) = 0.738112$ and $y_5 = 0.062614$, $P(8) = 0.605757$ and $y_8 = 0.064664$; Q6.1. $P(1) = 0.950001$, $P(2) = 0.900007$, $P(3) = 0.909996$; Q6.2. $F(0, 1, 2) = 5.555\%$, $F(0, 1, 3) = 2.174\%$, $F(0, 2, 3) = -1.098\%$.